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Precautions

- Never vortex cells
- Perform in accordance with BSL2
- Review risk assessment related to Trypan Blue and Turk

Reagents

- RPMI 1640 Glutamax ([Gibco, 72400-021](#))
- FBS ([Gibco, 1000500-064](#))
- dPBS [-] Ca2+ [-] Mg2+ ([Sigma Aldrich, D8537](#))
- Heparine 1000 IE ([LEO Pharma, 585679](#))
- Lymphoprep (Alere Technologies A/S, 1114547)
- DMSO ([Sigma Aldrich, D2650](#))
- Turk Gentiana violet (Histolab, 3605-B)
- Trypan blue, ([Sigma Aldrich, T8154-20ML](#))

Plastic

- 50 ml tubes red cap ([Sarstedt, 62557254](#))
- 15 ml tube red cap ([Sarstedt, 62554502](#))
- 1.8 ml cryovial ([Thermo Scientific, 377267](#))

Protocol

1. BM is diluted at the clinic in 3 ml RPMI1640 with 10 % FBS and 0.05 ml Heparine.
2. Note ml BM received
3. Into 3 ml Lymphoprep aliquot tubes marked with patient ID, *slowly* pipette down 3-4 ml BM, careful not to mix the two.
4. Centrifuge (400 g, 30 min, RT, acc 8, dec 3).
5. Carefully aspirate the MNC layer and add into 2 x 15 ml falcon tubes.
6. Wash with 5 ml wash media (dPBS + 10% FBS), centrifuge (300 g, 10 min, RT, 9 acc, 9 dec).
7. Pour out the wash media, flick to mix and then add another 5 ml wash media on top, centrifuge for another 10 minutes.
8. Dissolve MNCs in 5 ml wash buffer and pool in one 15 ml falcon tube.

9. Calculate MNCs per ml, total number of MNCs and % viability, with Turk (380µL + 20µL) or Trypan blue if sample is more than 24h old. Cells per mL = cells per large square * 20 (dilution factor) * 10 000 (volume factor).
10. Centrifuge (300 g, 10 min, RT, 9 acc, 9 dec).
11. Create aliquots according to table below. Dissolve correct number of MNCs in freezing media 1 (80% RPMI, 20% FBS), then add freezing media 1 for a total of 1000 µl freezing media 1 + MNCs in each cryovial. Then, add 1000 µL freezing media 2 (80% FBS, 20% DMSO) to each cryovial and immediately put the vials into a foam freezing chamber in a -80°C fridge over night.
12. After 24h in -80°C fridge, move the cells to -140°C biobank.

# BM MNCs	Aliquots
≤ 15 M	3 x 5 M
15 – 20 M	2 x 5 M and 1 x 10 M
20 – 100 M	2 x 5 M, 2 x 10 M, 3 x 20 M
> 100 M	2 x 5 M, 2 x 10 M, 10 x 20 M, the rest 50 M

References

1. "HERM MDS Biobank BM sampling Version 6 2025 02 14" by Indira Barbosa
2. "SOP MDS BIOBANK MEDIA_221003_version 1" by Indira Barbosa and Ann-Charlotte Björklund
3. "KI_SOP_TC_Freezing and thawing cells" by Kari Högstrand and Ellen Markljung